

DAVWO / ANI(TM) -- API Connections Report

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Repository: DavwoEnergyLtd/davwo-mvp1

Live preview: davwo-mvp1 (Vercel, production)

This document lists every API the platform connects to -- external third-party data/service providers, and the platform's own internal API surface -- grouped by purpose, with live production configuration status and, for anything not yet working, exact setup steps.

1. External APIs -- status at a glance

API	Category	Status	Env var(s)
UK Carbon Intensity API (National Grid ESO)	Energy & Grid Data	Live	none (keyless)
Octopus Energy Agile API	Energy & Grid Data	Live	none (keyless)
EIA (US Energy Information Administration)	Energy & Grid Data	Not configured	EIA_API_KEY
ENTSO-E Transparency Platform	Energy & Grid Data	Not built + not configured	ENTSOE_API_TOKEN
Open-Meteo	Weather	Live	none (keyless)
OpenChargeMap	EV Charge-Point Data	Not configured	OCM_API_KEY
OpenAI (Chat Completions)	AI / LLM Narration	Live	OPENAI_API_KEY
Resend	Email / Notifications	Not configured	RESEND_API_KEY

Every connector degrades gracefully with no key set -- a missing key never crashes the app, it just means that specific feature falls back to synthetic/no-op behaviour until configured. "Not configured" below means exactly that: the code is real and tested, it just has nothing to call yet in production.

2. Energy & Grid Data APIs

Per-region connectors live in `src/lib/data/regions/*.ts` -- an organisation's market field (`uk / eu / us`) selects which set of these `providers.ts` calls.

UK Carbon Intensity API (National Grid ESO) -- Live

- **Endpoint:** `api.carbonintensity.org.uk (/intensity, /generation, /intensity/{date}/fw24h)`
- **Purpose:** Live national grid carbon intensity (gCO2/kWh) and generation fuel mix -- powers the carbon gauge, the generation-mix chart, the "CO2 avoided" metric, and the carbon half of the smart-charging window.
- **Auth:** None required -- free, public, no key.
- **Used by:** `src/lib/data/regions/uk.ts`

Octopus Energy Agile API -- Live

- **Endpoint:** `api.octopus.energy/v1/products/{product}/electricity-tariffs/{tariff}/standard-unit-rates/`

- **Purpose:** Live half-hourly UK electricity pricing -- powers average cost/kWh, the price curve, cost-optimisation savings, and the price half of the smart-charging window.
- **Auth:** None required -- free, public, no key.
- **Used by:** `src/lib/data/regions/uk.ts`

EIA -- US Energy Information Administration -- Not configured

- **Endpoint:** `api.eia.gov/v2/` (`electricity/rto/fuel-type-data`, `electricity/retail-sales`)
- **Purpose:** Generation-mix-derived carbon intensity (EIA has no ready-made single figure like the UK's, so it's calculated from published fuel-mix data using standard lifecycle emission factors -- disclosed as a derived estimate, not a metered number) and monthly average retail electricity pricing, for organisations with `market: "us"`.
- **Current behaviour without a key:** `fetchCarbonIntensity()` and `fetchPricing()` both return `null` -- a `market: "us"` org sees synthetic/fallback data, not real US numbers.
- **Fix -- 5 minutes, no waiting period:**
 1. Register at <https://www.eia.gov/opendata/register.php> (instant, free, no approval wait).
 2. Add `EIA_API_KEY=<your key>` in Vercel -> Project -> Settings -> Environment Variables (Production), and in `.env.local` for local dev.
 3. Redeploy (`vercel --prod`) -- no code changes needed.
- Optional overrides (defaults are already sensible): `EIA_RESPONDENT` (default US48, national aggregate), `EIA_RETAIL_STATE` (default US), `EIA_RETAIL_SECTOR` (default RES, residential).

ENTSO-E Transparency Platform -- Not built + not configured

- **Endpoint:** ENTSO-E Transparency Platform REST API (day-ahead prices + generation mix, per European bidding zone)
- **Purpose:** The EU-market equivalent of the UK's carbon/pricing connectors -- real day-ahead price and generation-mix data for European organisations (`market: "eu"`).
- **Current behaviour:** Two things are missing, not one -- there is no `src/lib/data/regions/eu.ts` module yet (an EU-market org's requests are routed to the UK data set as a placeholder fallback), **and** no token is configured even if the module existed.
- **Fix:**
 1. Create an account at <https://transparency.entsoe.eu>.
 2. Email transparency@entsoe.eu with subject "RESTful API access" requesting a personal API token -- this is a manual approval step, **~3 business days**, and only Nathan/Collins can request it (it needs a real ENTSO-E account holder).
 3. Once the token arrives, set `ENTSOE_API_TOKEN` in Vercel Production + `.env.local`.
 4. Build `src/lib/data/regions/eu.ts` (mirrors `us.ts`'s structure -- this is a coding task, separate from the token request, and doesn't need to wait for the token to be written, only to be live-verified).
- **Action needed now, independent of any code work:** send the token request email if it hasn't been sent yet -- it's the single longest lead-time item in this whole report.

3. Weather

Open-Meteo -- Live

- **Endpoint:** `api.open-meteo.com/v1/forecast`

- **Purpose:** Temperature range and peak solar irradiance -- feeds the demand-forecast "drivers" shown on /forecasting and in ANI's forecast answers. Global coverage, so both the UK and US region modules already share this one connector with their own coordinates.
 - **Auth:** None required -- free, public, no key.
 - **Used by:** `src/lib/data/weather.ts` (shared by `regions/uk.ts` and `regions/us.ts`)
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4. EV Charge-Point / Location Data

OpenChargeMap -- Not configured

- **Endpoint:** `api.openchargemap.io/v3/poi/`
 - **Purpose:** Real EV charge-point locations and operational status for the Network Map (/map) -- falls back to the synthetic engine's own station pins when unavailable.
 - **Current behaviour without a key:** `fetchChargePoints()` returns `null` unconditionally (unlike the keyless APIs above, this one is coded to require a key, not just work at a reduced rate limit) -- every org, UK or US, currently sees the synthetic station map, not real nearby charge points.
 - **Fix -- free, self-service, a few minutes:**
 1. Register at <https://openchargemap.org> and generate an API key from your account.
 2. Add `OCM_API_KEY=<your key>` in Vercel Production + `.env.local`.
 3. Redeploy -- no code changes needed. Works immediately for both UK (`countrycode=GB`) and US (`countrycode=US`) organisations, since it's one shared connector.
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5. AI / LLM Narration

OpenAI (Chat Completions API) -- Live

- **Endpoint:** `api.openai.com/v1/chat/completions`
 - **Purpose:** Adds natural-language narration over ANI's already-verified data blocks (dashboard summaries, forecast explanations, the forecast-accuracy answer, product Q&A) -- it narrates and selects, it never invents the underlying numbers. ANI is fully functional without it: unset, every capability still answers with deterministic, keyword-routed template text.
 - **Auth:** `OPENAI_API_KEY`, currently set in Vercel Production.
 - **Model:** `OPENAI_MODEL` (default `gpt-5.6-terra`, OpenAI's mid-tier "balanced" model) -- not currently overridden in Production, so it's running on the default. Worth a quick check against the OpenAI dashboard that this model id is still correct/available, since it was picked after this assistant's own knowledge cutoff.
 - **Used by:** `src/lib/server/llm.ts` (single shared client -- replaced three near-duplicate call sites across `assistant.ts` and `productQA.ts`).
 - **Note:** the platform previously used Anthropic's Claude for this same role; it was fully migrated to OpenAI and no Anthropic code paths remain.
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6. Email / Notifications

Resend -- Not configured

- **Purpose:** All transactional/notification email -- procurement-enquiry notifications, invite-flow credentials, proactive asset-fault alerts, the on-demand digest, and scheduled PDF reports (the daily cron job).

- **Current behaviour without a key:** Every email path runs in a documented graceful no-op mode -- built, tested, and silently doing nothing in production. No crashes, no queued/lost mail, just nothing sent.
 - **Fix:**
 1. Create an account at <https://resend.com> (only Nathan or Collins can do this -- it's an account-creation step, not something achievable via CLI/code access).
 2. Generate an API key, add `RESEND_API_KEY=<your key>` in Vercel Production + `.env.local`.
 3. Optionally verify a sending domain and set `EMAIL_FROM=DAVWO ANI <alerts@yourdomain.com>` -- without this it defaults to Resend's shared test sender (`onboarding@resend.dev`), which only delivers to the Resend account's own email address, not real customers.
 - **This is the platform's single most-widely-blocked feature** -- five separate email flows are all waiting on this one key.
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7. Internal Platform API Surface

Everything below is the platform's own code -- no external dependency, all live and working in production today. Grouped by function, not exhaustively described per-endpoint (65 routes in total). All paths are relative to `/api/`.

Authentication & Identity -- customer sign-in, one-click demo roles, session bootstrap, self-service signup, admin org-switching.

`auth/login`, `auth/demo-login`, `auth/me`, `auth/signup`, `auth/switch-org`

Supplier Authentication -- a fully separate account/JWT model from customer auth, so a supplier session can never be confused with a customer session.

`supplier/auth/login`, `supplier/auth/me`, `supplier/auth/signup`

Dashboard & Operations -- headline KPIs, time-series charts, live station status/faults, cost/carbon optimisation, demand forecasting.

`dashboard/metrics`, `dashboard/energy-series`, `dashboard/recommendation`, `dashboard/map`, `monitoring`, `monitoring/faults`, `optimisation`, `forecasting`

Grid Data -- direct pass-throughs of the live carbon/pricing data from Section 2, shaped for specific charts/widgets.

`grid/carbon`, `grid/generation-mix`, `grid/pricing`, `grid/price-curve`, `grid/smart-window`

ANI(TM) Intelligence -- natural-language Q&A, prioritised explainable findings, proactive daily briefing, backtested forecast-accuracy reporting.

`ani/chat`, `ani/insights`, `ani/briefing`, `ani/forecast-accuracy`

Assets -- asset register (register/edit/delete), single + CSV-batch telemetry ingestion.

`assets`, `assets/[id]`, `assets/[id]/readings`, `assets/import`, `assets/readings/import`

Alerts -- operational alert feed with per-org acknowledge/resolve state.

`alerts`, `alerts/[id]/acknowledge`, `alerts/[id]/resolve`

Marketplace -- Customer side -- vendor/product discovery, AI-grounded product Q&A, enquiry capture, procurement state (new -> responded -> accepted/declined).

`marketplace/vendors`, `marketplace/vendors/[id]`, `marketplace/products`, `marketplace/products/[id]`, `marketplace/contact`, `marketplace/ask`, `marketplace/leads`, `marketplace/leads/[id]`

Marketplace -- Supplier side -- supplier self-service: product catalogue, technical documents, per-product knowledgebase, lead management.

marketplace/suppliers, marketplace/suppliers/[id], marketplace/products/[id]/documents (+[docId]), marketplace/products/[id]/knowledgebase, supplier/products, supplier/products/[id]/documents, supplier/products/[id]/knowledgebase, supplier/leads, supplier/leads/[id]

Analytics & Reports -- period-over-period comparison analytics, PDF/CSV report generation.

analytics/summary, reports/export

Organisation, Users & Settings -- org profile, team management, role changes, invites, password changes, per-user preferences.

org, tenants, users, users/[id], users/invite, users/change-password, users/preferences

Notifications -- in-app notification feed, on-demand email digest.

notifications, notifications/email

Pilot Onboarding -- ROI-estimate generation from a stated infrastructure profile.

pilot/onboard

Ops / Infrastructure -- liveness check for uptime monitoring; daily scheduled report emails (guarded by CRON_SECRET).

health, cron/reports

8. Summary -- what needs action, and by whom

Item	Who	Effort	Blocked on
Set EIA_API_KEY	Nathan/Collins	~5 min	Nothing -- instant registration
Set OCM_API_KEY	Nathan/Collins	~5 min	Nothing -- instant registration
Set RESEND_API_KEY (+ EMAIL_FROM)	Nathan/Collins	~10 min	Account creation only
Send the ENTSO-E token request	Nathan/Collins	~5 min to send	~3 business days for approval to arrive
Build src/lib/data/regions/eu.ts	Development	Coding task	Can start now, independent of the token

Every one of these is additive -- none of them require any change to already-shipped, already-tested code paths. Setting a key simply switches that feature from its graceful synthetic/no-op fallback to live data.